

FURTHER MECHANICS / MISCELLANEOUS MECHANICS

Dimensional Analysis

Worked solutions

10 questions · 53 marks

Instructions

Unless a question states otherwise, take $g = 9.8 \text{ m s}^{-2}$ wherever a numerical value of g is required. Solutions also give final answers for $g = 10 \text{ m s}^{-2}$ where these differ.

Questions

Question 1	5 marks	2
Question 2	4 marks	3
Question 3	4 marks	4
Question 4	4 marks	6
Question 5	7 marks	7
Question 6	6 marks	9
Question 7	4 marks	10
Question 8	6 marks	11
Question 9	7 marks	12
Question 10	6 marks	14

1. (a) State the dimensions of force.

[1]

(b) The braking distance of a vehicle can be modelled by the formula

$$d = m^a u^b R^c$$

where,

m is the mass of the vehicle in kg

u is the initial speed in m s^{-1}

R is the constant braking force in N

Use dimensional analysis to find the values of a , b and c .

[4]

Solution

(a) Force is mass $\times$ acceleration, so

$$[F] = M \times LT^{-2} = MLT^{-2}$$

(b) Write the dimensions of each quantity:

$$[d] = L, \quad [m] = M, \quad [u] = LT^{-1}, \quad [R] = MLT^{-2}$$

Given

$$d = m^a u^b R^c$$

the dimensions on the right-hand side are

$$\begin{aligned} [m^a u^b R^c] &= M^a (LT^{-1})^b (MLT^{-2})^c \\ &= M^a L^b T^{-b} \times M^c L^c T^{-2c} \\ &= M^{a+c} L^{b+c} T^{-b-2c} \end{aligned}$$

Since this must equal the dimensions of d ,

$$M^{a+c} L^{b+c} T^{-b-2c} = L = M^0 L^1 T^0$$

Equating powers of M , L and T gives

$$a + c = 0, \quad b + c = 1, \quad -b - 2c = 0$$

From $-b - 2c = 0$,

$$b = -2c$$

Substitute into $b + c = 1$:

$$-2c + c = 1$$

$$-c = 1$$

$$c = -1$$

Then

$$b = -2(-1) = 2$$

and from $a + c = 0$,

$$a + (-1) = 0$$

so

$$a = 1$$

Hence

$$a = 1, \quad b = 2, \quad c = -1$$

2. A small satellite moves in a circular orbit of radius r about the centre of a much more massive planet of mass M . It is suggested that the period T of the orbit depends only on G , M and r , and that

$$T = kG^\alpha M^\beta r^\gamma$$

where k is a dimensionless constant.

It is given that

$$G = 6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$$

(a) Write down the dimensions of G . [1]

(b) By using dimensional analysis, determine the values of α , β and γ . [3]

Solution

(a) From

$$G = 6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$$

the dimensions of G are

$$[G] = M^{-1}L^3T^{-2}$$

(b) Since k is dimensionless,

$$[T] = [G]^\alpha [M]^\beta [r]^\gamma$$

Now

$$[T] = T, \quad [G] = M^{-1}L^3T^{-2}, \quad [r] = L$$

so

$$\begin{aligned} T &= (M^{-1}L^3T^{-2})^\alpha M^\beta L^\gamma \\ &= M^{-\alpha+\beta} L^{3\alpha+\gamma} T^{-2\alpha} \end{aligned}$$

Equating powers of M , L and T with $M^0L^0T^1$:

$$-\alpha + \beta = 0$$

$$3\alpha + \gamma = 0$$

$$-2\alpha = 1$$

From $-2\alpha = 1$,

$$\alpha = -\frac{1}{2}$$

Then

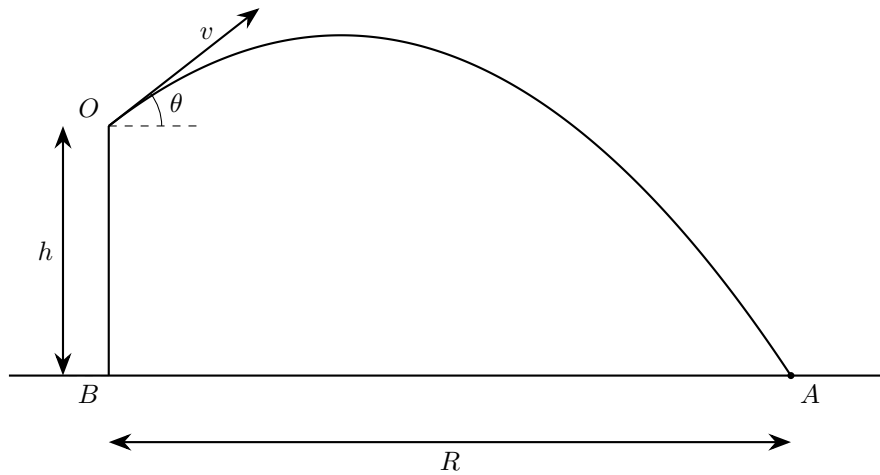
$$\beta = \alpha = -\frac{1}{2}$$

and

$$\gamma = -3\alpha = \frac{3}{2}$$

Hence

$$\alpha = -\frac{1}{2}, \quad \beta = -\frac{1}{2}, \quad \gamma = \frac{3}{2}$$



3. A particle is projected with speed v from a point O at the top of a cliff of height h above horizontal ground. The angle of projection is θ above the horizontal. The particle lands at the point A on the ground, and B is the foot of the cliff, so that $BA = R$, as shown in the diagram.

You are given that

$$R = \frac{v^\alpha \cos \theta}{g^\beta} \left(v^\gamma \sin \theta + \sqrt{v^\delta \sin^2 \theta + 2gh} \right)$$

Use dimensional analysis to find α , β , γ and δ .

[4]

Solution

Using dimensions,

$$[R] = L, \quad [v] = LT^{-1}, \quad [g] = LT^{-2}, \quad [h] = L$$

and $\sin \theta$ and $\cos \theta$ are dimensionless.

First look at the expression inside the square root. Since terms being added must have the same dimensions,

$$[v^\delta \sin^2 \theta] = [2gh]$$

Now

$$[2gh] = [g][h] = (LT^{-2})(L) = L^2T^{-2}$$

So

$$(LT^{-1})^\delta = L^2T^{-2}$$

Comparing powers of L and T ,

$$\delta = 2$$

Hence the whole square-root term has dimensions

$$\sqrt{L^2T^{-2}} = LT^{-1}$$

Therefore $v^\gamma \sin \theta$ must also have dimensions LT^{-1} , so

$$(LT^{-1})^\gamma = LT^{-1}$$

which gives

$$\gamma = 1$$

Now the bracket

$$v^\gamma \sin \theta + \sqrt{v^\delta \sin^2 \theta + 2gh}$$

has dimensions LT^{-1} . So

$$\begin{aligned}[R] &= \left[\frac{v^\alpha \cos \theta}{g^\beta} \right] \left[v^\gamma \sin \theta + \sqrt{v^\delta \sin^2 \theta + 2gh} \right] \\ L &= \frac{(LT^{-1})^\alpha}{(LT^{-2})^\beta} (LT^{-1}) \\ &= L^{\alpha-\beta+1} T^{-\alpha+2\beta-1}\end{aligned}$$

Comparing powers with $L^1 T^0$,

$$\alpha - \beta + 1 = 1, \quad -\alpha + 2\beta - 1 = 0$$

From the first equation,

$$\alpha = \beta$$

Substitute into the second:

$$-\beta + 2\beta - 1 = 0$$

so

$$\beta = 1$$

and therefore

$$\alpha = 1$$

So the values are

$$\alpha = 1, \quad \beta = 1, \quad \gamma = 1, \quad \delta = 2$$

4. The increase in pressure p pascals between the surface and the bottom of a liquid of depth h metres and density $\rho \text{ kg m}^{-3}$ is modelled by

$$p = k\rho^\alpha h^\beta g$$

where $g \text{ m s}^{-2}$ is the acceleration due to gravity and k is a dimensionless constant.

- (a) State the dimensions of p . [1]
- (b) Use dimensional analysis to find the value of α and the value of β . [3]

Solution

- (a) Pressure is force per unit area.

Since $[\text{force}] = MLT^{-2}$ and $[\text{area}] = L^2$,

$$[p] = \frac{MLT^{-2}}{L^2} = ML^{-1}T^{-2}$$

- (b) Using

$$p = k\rho^\alpha h^\beta g$$

and noting that k is dimensionless,

$$[p] = [\rho]^\alpha [h]^\beta [g]$$

Now

$$[\rho] = ML^{-3}, \quad [h] = L, \quad [g] = LT^{-2}$$

So the dimensions of the right-hand side are

$$\begin{aligned} (ML^{-3})^\alpha (L)^\beta (LT^{-2}) &= M^\alpha L^{-3\alpha} L^\beta LT^{-2} \\ &= M^\alpha L^{-3\alpha+\beta+1} T^{-2} \end{aligned}$$

But from part (a),

$$[p] = ML^{-1}T^{-2}$$

Equating powers of M , L and T :

$$\alpha = 1$$

and

$$-3\alpha + \beta + 1 = -1$$

Substituting $\alpha = 1$,

$$\begin{aligned} -3(1) + \beta + 1 &= -1 \\ \beta - 2 &= -1 \\ \beta &= 1 \end{aligned}$$

Hence

$$\alpha = 1, \quad \beta = 1$$

5. The force, F , exerted on a flat plate by a water jet is thought to depend on the density of the water, ρ , the cross-sectional area of the jet, A , and the speed of the water, v . You are given that ρ is measured in kg m^{-3} .

- (a) Find the dimensions of density ρ . [2]

A model of the force suggests the relationship

$$F = \frac{1}{2} \rho^\alpha A^\beta v^\gamma$$

- (b) Explain whether or not it is possible to use dimensional analysis to verify that the constant $\frac{1}{2}$ is correct. [1]

- (c) Use dimensional analysis to find the values of α , β and γ . [4]

Solution

- (a) Density is mass per unit volume.

$$[\rho] = \frac{M}{L^3} = ML^{-3}$$

- (b) No, it is not possible.

Dimensional analysis only checks that both sides of an equation have the same dimensions. The number $\frac{1}{2}$ is a dimensionless constant, so dimensional analysis cannot show whether $\frac{1}{2}$ is the correct numerical value.

- (c) We use

$$F = \frac{1}{2} \rho^\alpha A^\beta v^\gamma$$

The factor $\frac{1}{2}$ is dimensionless, so it does not affect the dimensions.

Now write the dimensions of each quantity:

$$[F] = MLT^{-2}, \quad [\rho] = ML^{-3}, \quad [A] = L^2, \quad [v] = LT^{-1}$$

So

$$MLT^{-2} = (ML^{-3})^\alpha (L^2)^\beta (LT^{-1})^\gamma$$

Combining powers gives

$$MLT^{-2} = M^\alpha L^{-3\alpha+2\beta+\gamma} T^{-\gamma}$$

Equating powers of M , L and T :

$$\text{For } M : \quad \alpha = 1$$

$$\text{For } T : \quad -\gamma = -2 \Rightarrow \gamma = 2$$

$$\text{For } L : \quad -3\alpha + 2\beta + \gamma = 1$$

Substitute $\alpha = 1$ and $\gamma = 2$ into the L -equation:

$$-3(1) + 2\beta + 2 = 1$$

$$-1 + 2\beta = 1$$

$$2\beta = 2$$

$$\beta = 1$$

Hence

$$\alpha = 1, \quad \beta = 1, \quad \gamma = 2$$

So the model is

$$F = \frac{1}{2}\rho Av^2$$

6. A motor pulls a crate of mass m along a rough horizontal floor. The equation below is suggested to model the pulling force exerted by the motor, F , while the crate moves in a straight line.

$$F = k_1 m v \frac{dv}{dx} + k_2 \mu m g + k_3 \frac{Q}{x}$$

in which,

- v is the speed of the crate when it has travelled distance x
- μ is the coefficient of friction between the crate and the floor
- g is the acceleration due to gravity
- Q is the total thermal energy transferred to the surroundings after the crate has moved distance x
- k_1 , k_2 and k_3 are dimensionless constants

Determine whether the equation is dimensionally consistent.

[6]

Solution

For the equation to be dimensionally consistent, every term must have the dimensions of force.

$$[F] = MLT^{-2}$$

Also, k_1 , k_2 and k_3 are dimensionless, and μ is dimensionless.

Now consider each term on the right-hand side.

$$[v] = LT^{-1}, \quad [x] = L$$

so

$$\left[\frac{dv}{dx} \right] = \frac{[v]}{[x]} = \frac{LT^{-1}}{L} = T^{-1}$$

Hence for the first term,

$$\begin{aligned} \left[k_1 m v \frac{dv}{dx} \right] &= [m][v] \left[\frac{dv}{dx} \right] \\ &= M \cdot LT^{-1} \cdot T^{-1} \\ &= MLT^{-2} \end{aligned}$$

For the second term,

$$\begin{aligned} [k_2 \mu m g] &= [\mu][m][g] \\ &= 1 \cdot M \cdot LT^{-2} \\ &= MLT^{-2} \end{aligned}$$

For the third term, Q is thermal energy transferred, so it has the dimensions of energy:

$$[Q] = ML^2T^{-2}$$

Therefore

$$\begin{aligned} \left[k_3 \frac{Q}{x} \right] &= \frac{[Q]}{[x]} \\ &= \frac{ML^2T^{-2}}{L} \\ &= MLT^{-2} \end{aligned}$$

Each term on the right-hand side has dimensions MLT^{-2} , the same as force. Therefore the equation is dimensionally consistent.

7. The speed of a particle falling vertically through a resistive medium is modelled by the formula

$$v = \frac{mg}{k} \left(1 - e^{-kt/m}\right)$$

where,

v is the speed of the particle

m is the mass of the particle

g is the acceleration due to gravity

t is the time since the particle was released

k is a constant

Use dimensional analysis to determine the dimensions of k .

[4]

Solution

For

$$v = \frac{mg}{k} \left(1 - e^{-kt/m}\right)$$

the factor $1 - e^{-kt/m}$ must be dimensionless.

In particular, the exponent of an exponential must have no dimensions, so

$$\left[\frac{kt}{m}\right] = 1$$

Using

$$[t] = T, \quad [m] = M$$

we get

$$\frac{[k][t]}{[m]} = 1$$

so

$$\frac{[k]T}{M} = 1$$

Hence

$$[k] = MT^{-1}$$

As a check,

$$\left[\frac{mg}{k}\right] = \frac{M \cdot LT^{-2}}{MT^{-1}} = LT^{-1}$$

which is the dimension of speed, as required.

Therefore k has dimensions MT^{-1} , with SI unit kg s^{-1} .

8. A comet of mass m measured in kg follows an elliptical orbit around a planet. The semi-major axis of the orbit is a measured in metres and the orbital period is T measured in seconds. The eccentricity e of the orbit is dimensionless.

A constant μ associated with the planet is defined by

$$T^2 = \frac{4\pi^2 a^3}{\mu}$$

and the angular momentum of the comet is given by

$$H = m\sqrt{\mu a(1 - e^2)}$$

By using dimensional analysis, find the dimensions of:

(a) μ

[3]

(b) H

[3]

Solution

(a) From

$$T^2 = \frac{4\pi^2 a^3}{\mu}$$

and since $4\pi^2$ is dimensionless,

$$[T]^2 = \frac{[a]^3}{[\mu]}$$

Using $[T] = T$ and $[a] = L$,

$$T^2 = \frac{L^3}{[\mu]}$$

so

$$[\mu] = \frac{L^3}{T^2} = L^3 T^{-2}$$

(b) The angular momentum is

$$H = m\sqrt{\mu a(1 - e^2)}$$

Here $[m] = M$, and since e is dimensionless, $(1 - e^2)$ is also dimensionless.

Therefore

$$[H] = [m]\sqrt{[\mu][a]}$$

Using $[\mu] = L^3 T^{-2}$ from part (a) and $[a] = L$,

$$\begin{aligned} [H] &= M\sqrt{(L^3 T^{-2})(L)} \\ &= M\sqrt{L^4 T^{-2}} \\ &= ML^2 T^{-1} \end{aligned}$$

9. A small ball of mass m kg is projected vertically upwards with speed u m s⁻¹. During the motion, when its speed is v m s⁻¹, it experiences an air resistance of magnitude kv^2 N, where k is a constant.

(a) Determine the dimensions of k .

[3]

The greatest height reached by the ball is H m. A formula approximating H is

$$H = \frac{u^2}{2g} - \frac{ku^4}{4mg^2} + \frac{1}{6}k^2m^\alpha u^\beta g^\gamma$$

where g m s⁻² is the acceleration due to gravity.

(b) Use dimensional analysis to find α , β and γ .

[4]

Solution

(a) The air resistance has magnitude kv^2 , and this is a force.

$$[k][v]^2 = [F]$$

Now

$$[F] = MLT^{-2}, \quad [v] = LT^{-1}$$

So

$$\begin{aligned} [k] &= \frac{[F]}{[v]^2} \\ &= \frac{MLT^{-2}}{(LT^{-1})^2} \\ &= \frac{MLT^{-2}}{L^2T^{-2}} \\ &= ML^{-1} \end{aligned}$$

Hence the SI unit of k is kg m⁻¹.

(b) Since H is a height,

$$[H] = L$$

Therefore each term in

$$H = \frac{u^2}{2g} - \frac{ku^4}{4mg^2} + \frac{1}{6}k^2m^\alpha u^\beta g^\gamma$$

must have dimensions of length.

So the third term must satisfy

$$[k^2m^\alpha u^\beta g^\gamma] = L$$

Using

$$[k] = ML^{-1}, \quad [m] = M, \quad [u] = LT^{-1}, \quad [g] = LT^{-2}$$

we get

$$\begin{aligned} [k^2m^\alpha u^\beta g^\gamma] &= (ML^{-1})^2(M)^\alpha(LT^{-1})^\beta(LT^{-2})^\gamma \\ &= M^{2+\alpha}L^{-2+\beta+\gamma}T^{-\beta-2\gamma} \end{aligned}$$

This must equal $M^0L^1T^0$, so equating powers gives

$$\begin{aligned} 2 + \alpha &= 0 \\ -2 + \beta + \gamma &= 1 \\ -\beta - 2\gamma &= 0 \end{aligned}$$

From the first equation,

$$\alpha = -2$$

From the third equation,

$$\beta = -2\gamma$$

Substitute into the second equation:

$$-2 + (-2\gamma) + \gamma = 1$$

$$-2 - \gamma = 1$$

$$-\gamma = 3$$

$$\gamma = -3$$

Then

$$\beta = -2(-3) = 6$$

Hence

$$\alpha = -2, \quad \beta = 6, \quad \gamma = -3$$

10. A small probe is launched vertically upwards from the surface of a spherical moon. The radius of the moon is R metres, and the probe is at a distance r metres from the centre of the moon. The gravitational field strength g at distance r is modelled by

$$g = \frac{\Gamma}{r^2}$$

where Γ is a constant associated with the moon.

The probe has initial speed u at the surface and speed v when it reaches distance r . These speeds are related by

$$v^2 = u^2 - 2\Gamma \left(\frac{1}{R} - \frac{1}{r} \right)$$

The work done against gravity in moving a probe of mass m from the surface to distance r is

$$W = m\Gamma \left(\frac{1}{R} - \frac{1}{r} \right)$$

- (a) Find the dimensions of Γ . [2]
 (b) Use dimensional analysis to show that the equation for v^2 is dimensionally consistent. [2]
 (c) Find the dimensions and SI units of W . [2]

Solution

- (a) From

$$g = \frac{\Gamma}{r^2}$$

and since

$$[g] = LT^{-2}, \quad [r] = L$$

we have

$$[\Gamma] = [g][r^2] = LT^{-2} \times L^2 = L^3T^{-2}$$

- (b) In

$$v^2 = u^2 - 2\Gamma \left(\frac{1}{R} - \frac{1}{r} \right)$$

the dimensions of v^2 and u^2 are

$$[v^2] = [u^2] = (LT^{-1})^2 = L^2T^{-2}$$

Also,

$$\left[\frac{1}{R} \right] = \left[\frac{1}{r} \right] = L^{-1}$$

so

$$\left[\left(\frac{1}{R} - \frac{1}{r} \right) \right] = L^{-1}$$

Since 2 is dimensionless,

$$\left[2\Gamma \left(\frac{1}{R} - \frac{1}{r} \right) \right] = [\Gamma]L^{-1} = L^3T^{-2} \times L^{-1} = L^2T^{-2}$$

Therefore every term in the equation has dimensions L^2T^{-2} , so the equation is dimensionally consistent.

- (c) Using

$$W = m\Gamma \left(\frac{1}{R} - \frac{1}{r} \right)$$

with

$$[m] = M, \quad [\Gamma] = L^3 T^{-2}, \quad \left[\left(\frac{1}{R} - \frac{1}{r} \right) \right] = L^{-1}$$

gives

$$[W] = M \times L^3 T^{-2} \times L^{-1} = M L^2 T^{-2}$$

Its SI units are

$$\text{kg m}^2 \text{s}^{-2}$$

which is the joule, J.